

Сортировки

$a = [2 \ 1 \ 3 \ 5 \ 4]$

$\text{sort}(a) = [1 \ 2 \ 3 \ 4 \ 5]$

`std::sort` - встроенная в C++ сортировка

Quick Sort + Insertion Sort

$O(n \log n)$, где n - размер массива

std::stable-sort

$a = [(1, 'a'), (1, 'b'), (2, 'a'), (1, 'c')]$

comparison not needed $(1, 'a') = (1, 'c')$

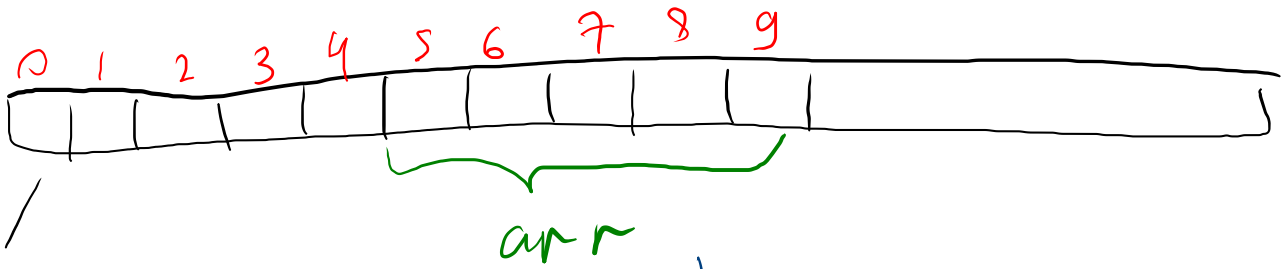
sort $\rightarrow$ $[1c, 1a, 1b, 2a]$

stable $\rightarrow$ $[1a, 1b, 1c, 2a]$

указатели

$\text{int}^* \text{arr} = 5$

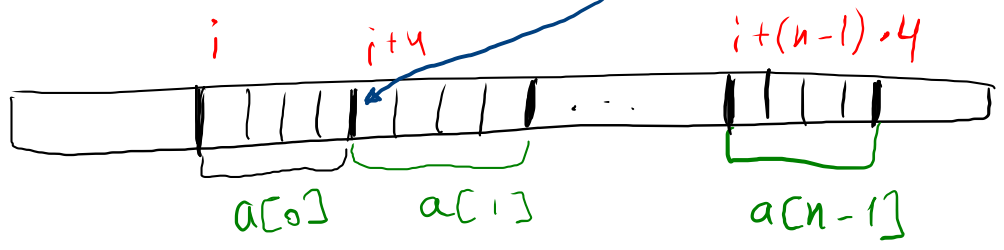
RAM



1 байт

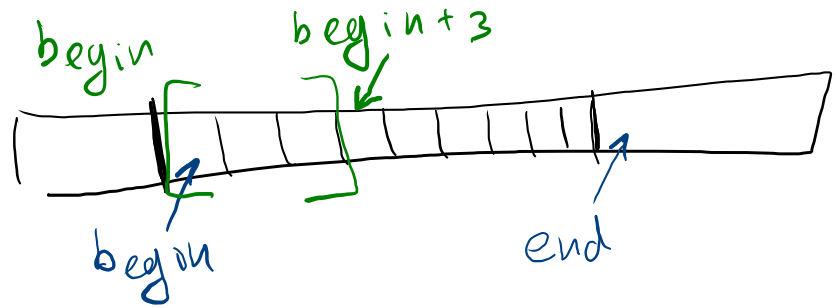
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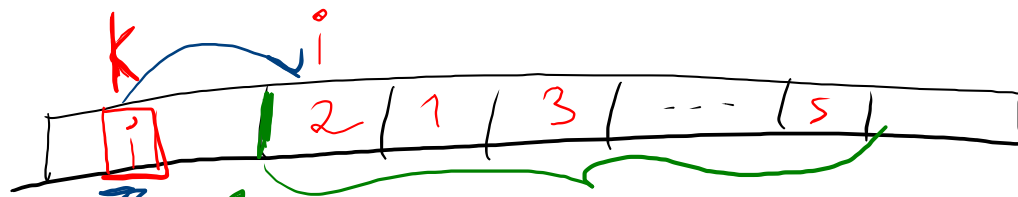
vector a



Итератор — указатель на элемент контейнера

$\text{iter} = i + 4$





vector

copy

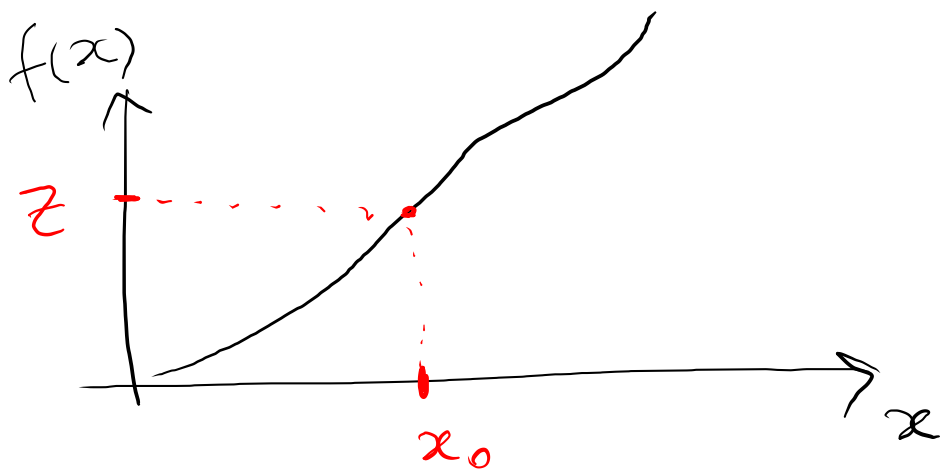


x

vector (8) x = k

agpec x

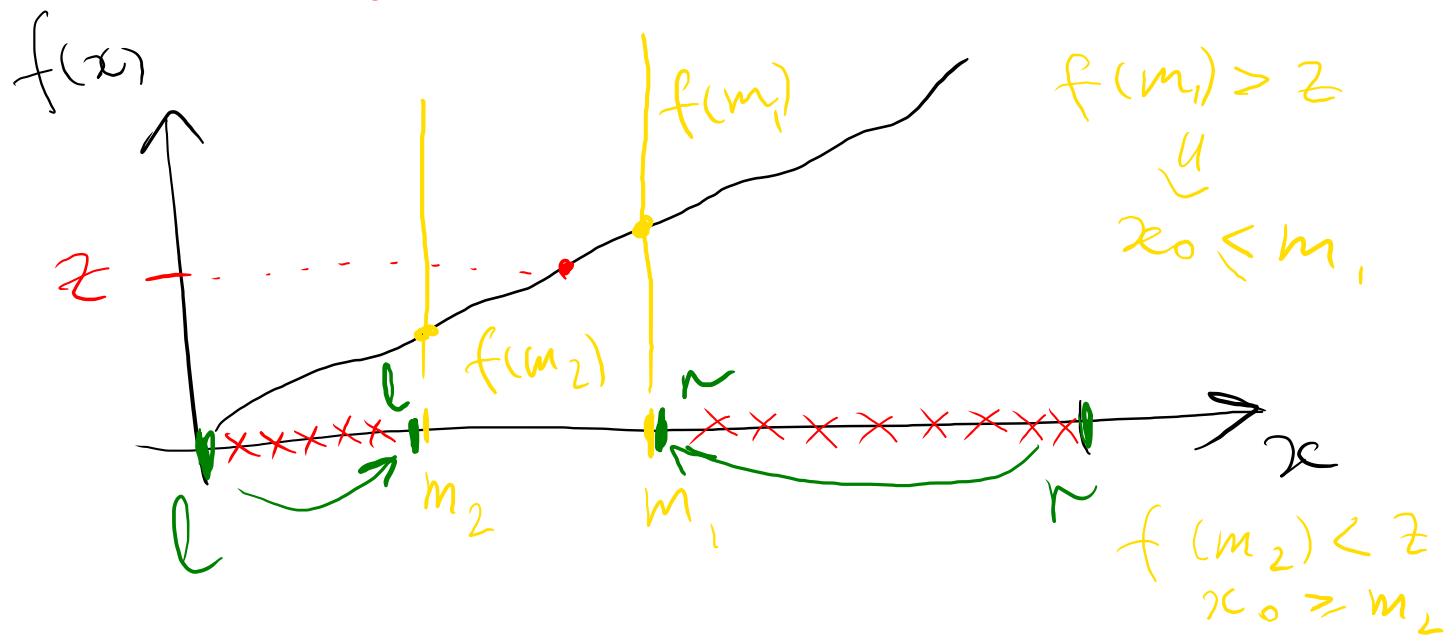
Бинарный поиск aka гуготомия



$$x \leq y \Rightarrow f(x) \leq f(y)$$

$$f(x_0) = z \quad x_0 = ?$$

$$O(\log(r-l))$$





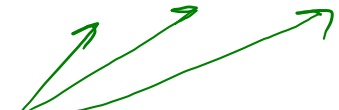
if ...
L2 m

Был поиск по массиву

$a = [2, 3, -1, 4, 7, 5]$

$[l, r)$

$a[l] = 5$

5? 

$O(q \cdot n)$
 l

m_1 m_2


sorted $a = [-1, 2, 3, 4, 5, 7]$




5? $4 \leq 5? \Rightarrow$

$a[m_1] \leq 5$

$l + 1 = r$ m_3

$a[m_2] \leq 5 \Rightarrow l = m$

$a[m_3] > 5 \Rightarrow r = m$

$O((q+n) \log n)$